

Group 1 :

A) Write the scientific term expressed by the following statements:

- 1) The magnetic force exerts on a wire of 1m length carrying a current of intensity 1 A and placed perpendicularly to the magnetic field.
- 2) The magnetic torque acting on a coil carrying current placed parallel to a uniform magnetic field of flux density of 1 Tesla.

B) Mention the factors affecting each of the following (one factor only)

- 3) The magnetic force acting on a wire carrying current placed perpendicular to a uniform magnetic field.
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- 4) The density of magnetic field at a point along interior axis of a solenoid carrying an electric current.

C) Put circle around the correct answer

- 5) A square of area of cross section 10^{-2} m^2 is placed with its axis perpendicular to a uniform magnetic field of flux density 1 T. Then the total magnetic flux through the coil is**
- a) 10^{-2} Wb b) $5 \times 10^{-3} \text{ Wb}$
- c) $5\sqrt{3} \times 10^{-3} \text{ Wb}$ d) $5 \times 10^{-2} \text{ Wb}$
- 6) two metallic rings A and B, If a ring A of radius r , current I is flowing and in ring B of radius 2r , current 2I is flowing, then**
- $\frac{\text{the density of magnetic field at center of ring A } (B_A)}{\text{the density of magnetic field at center of ring B } (B_B)} = \dots$**
- a) $\frac{1}{4}$ b) $\frac{1}{2}$ c) $\frac{1}{1}$ d) $\frac{4}{1}$

D) Answer the following question

- 7) An electric current of intensity 10 A is flowing in a long straight wire. Calculate

8)	the density of magnetic field at a distance 0.1m from the wire (Permeability of air = $4\pi \times 10^{-7}$ Wb /A .m)
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E) Compare between each of the following

9 10	Point of comparison	Fleming left hand rule	Ampere's right hand rule
	The use		

Group 2 :**A) Write the scientific term expressed by the following statements:**

11) In the transformer the ratio of output electric power to the input electric power.

.....

- 12)** A phenomenon in which an induced electromotive force and also an induced current are generated in the conductor by a changing magnetic field (magnetic flux)

B) Mention the factors affecting each of the following (one factor only)

- 13)** The coefficient of mutual inductance between two coils.

- 14)** The instantaneous value of induced emf generated in the dynamo.

C) Put circle around the correct answer

- 15) A straight wire 1m long is moving in a uniform magnetic field of flux density 0.1T at a uniform velocity 10m/s. an induced potential difference of 0.5V is generated between its terminals. The angle between the direction of the wire motion and the direction of the magnetic field is.

- a) 0° b) 30° c) 60° d) 90°

- 16) A coil which carries current I is wound on iron core. The self-induced electromotive force in the coil is not affected by.....

- a) Variation in coil current
b) Variation in voltage to the coil
c) Change of number of turns of coil
d) The resistance of coil wire

- 17) According to the Lenz's law,.....**

- The direction of the induced emf will be such that it helps the change in magnetic flux
- The direction of the induced current will be such that it opposes the change in magnetic flux.
- The magnitude of the induced electromotive force (emf) is directly proportional to the rate of change of flux.
- The direction of the induced electromotive force (emf) will be along the direction of the magnetic field.

D) Correct the underlined in the following statement

- 18)** A laminated iron core is used in an electric transformer to **increase** eddy currents.

- 19)** The ac generator is a device based on the principle of **magnetic torque**.

- 20)** The induced electromotive force (emf) generated in a coil increases to **double** if the rate of change of current in the same coil increases 4 times

Group 3:

A) Put sign (✓) in front of the correct statement and sign (X) in front of the incorrect statement of each of the following:

- 21)** At low temperatures, the wavelengths of the thermal radiation are mainly in the infrared region. ()

22) In Compton Effect the frequency of the scattered photon is greater than the frequency of the incident photon ()

23) The maximum frequency of the continuous radiation of X-rays does not depend on the potential difference between the cathode and anode. ()

B) Put circle around the correct answer

24) The process in which electrons are emitted from a metallic surface when heat given to it is called.....

a) Photoelectric effect

b) Thermionic emission

c) Compton effect

d) Black body radiation

25) The common feature of laser and (x) rays photons are that they.....

a) are coherent

b) are monochromatic

c) have the same speed

d) have the same energy

26) To increase the number of photoelectrons emitted per second when light falling on a metallic surface,

a) The frequency of incident photon increases

b) Kinetic energy of emitted electron increases

c) The wavelength of incident photon increases

d) The intensity of the radiation increases

C) Mention the factors affecting each of the following (one factor only)

27) The wavelength of the characteristic X- rays.

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28) The wave length at maximum radiation intensity in Planck's curve

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D) Write the mathematical law used to calculate

29) Momentum of photon

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30) Kinetic energy of emitted electron when light falling on a metallic surface.

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Group 4:

A) Write the scientific term expressed by the following statements:

31) The number of bonds broken per second will be equal to the number of bonds mended per second.

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32) The ratio between the collector current and the base current

B) Put circle around the correct answer

33) Doping a semiconductor with impurities of atoms creates an n-type material.

- a) trivalent b) tetravalent c) pentavalent d) divalent

34) When connecting the p-n Junction (diode) in reverse biasing, then the width of depletion layer.....

- a) increases b) decreases c) does not change d) becomes zero

35) Which of the following gates that if both inputs are low the output is high?

- a) OR gate b) AND gate c) NAND gate d) NOT gate

C) Give a reason for each of the following

36) At very low temperature, pure silicon crystal is an insulator

37) The thickness of the base of the bipolar junction transistor must be very small

D) Put sign (✓) in front of the correct statement or sign (X) in front of the incorrect statement of each of the following:

38) In an NPN transistor, the emitter is doped with a higher concentration of impurities than the collector. ()

39) The resistance of PN junction obeys Ohm's law. ()

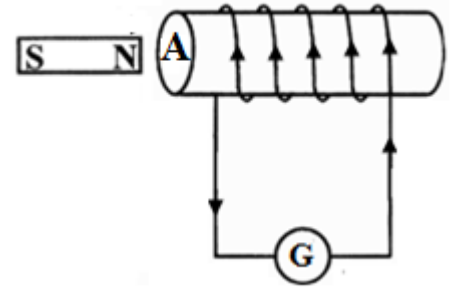
40) In NPN transistor, the collector current is 10mA. If 90% of the electrons emitted reach the collector, then the emitter current will be 11.3 mA ()

Group 5:

Answer the following

- | | |
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| 41) | A photon of wavelength 5.6×10^{-7} m falls on a metallic surface of work function 3.42×10^{-19} J, calculate the kinetic energy of emitted electron from the metallic surface .
(Planck's constant = 6.625×10^{-34} J.s, speed of light in air = 3×10^8 m/s) |
| 42) | |

	
43) 44)	A metallic ring is placed in a magnetic field which changes from 0.004 Wb to 0.008 Wb in 2s. Calculate the induced electromotive force(emf) in the coil
45) 46)	Mention without explanation the two conditions necessary for laser action
47) 48)	Mention only two factors affect The magnetic torque acting on a coil
49) 50)	In the opposite diagram a magnet is moving relative to the solenoid , look well at the opposite diagram then determine the following a) kind of the magnetic pole formed at A b) The direction of the magnet motion



(انتهت الاسئلة)